Midpoint & Distance Formula — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: The trick*

Key Idea

Midpoint = **M**__ean__ **D**istance = **D**__ifference__

☞ *Video: Midpoint: simple case first*

Key Idea



The middle of -3 and 3 is __0__.

Add and halve: $\dfrac{-3+3}{2} = 0$ ✓

So the midpoint __adds__ — it is the __average__.

☞ *Video: Midpoint formula*

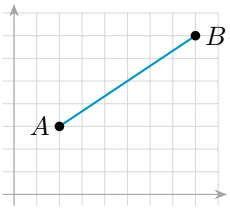
Key Idea

$M =$ $\left(\dfrac{x_1+x_2}{2}, \dfrac{y_1+y_2}{2}\right)$

Example: $A(2,3)$, $B(8,7)$

x: $\dfrac{2+8}{2} =$ __5__ y: $\dfrac{3+7}{2} =$ __5__

$M =$ $(5,5)$



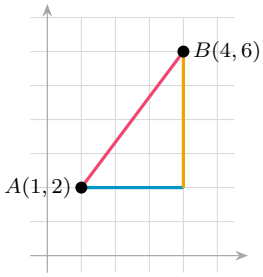
☞ *Video: Distance: simple case first*

Key Idea

From $(2,0)$ to $(5,0)$: count the squares: __3__. That is $5-2$ — a __difference__.

☞ *Video: Distance = Pythagoras on a grid*

Key Idea



Find AB: across __3__, up __4__.

$d = \sqrt{3^2+4^2} = \sqrt{25} =$ __5__

$d =$ $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

The distance formula is just the __Pythagorean Theorem__.

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☞ *Video: Two common mistakes*

Common Mistake

Square __each__ difference before adding: $\sqrt{3^2+4^2}=5$, not $3+4=$ __7__.

Testing add vs. subtract? Don't use 0 and 2 (both give 1). Use 2 and 4: the middle is __3__ $=\dfrac{2+4}{2}$.

☞ *Video: Your turn*

Your Turn

$P(-1,4)$ and $Q(5,-4)$. Find the midpoint of $\overline{PQ}$ and the length PQ.

$M=\left(\dfrac{-1+5}{2},\ \dfrac{4+(-4)}{2}\right)=$ __$(2,0)$__

$PQ=\sqrt{(5-(-1))^2+(-4-4)^2}=\sqrt{6^2+8^2}=\sqrt{100}=$ __10__

Notice

Remember: **Midpoint = Mean**: add the coordinates, divide by 2. **Distance = Difference**: subtract, square, add, square root.

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